

# L1.3 Continuity

§1.8 — what you need to know

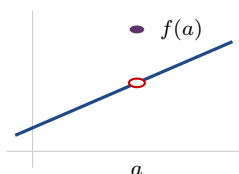
## CONTINUOUS AT $x = a$

$$\lim_{x \rightarrow a} f(x) = f(a)$$

**Expanded:**  $\text{LHL} = f(a) = \text{RHL}$

“which means you could draw the function without picking up your pen”

**Notation.** LHL RHL DNE · HE @ hole · Z @ zero · VA @ · HA @



the value is there, and *wrong*

## Kinds of discontinuity

**removable** (hole)

**the value** —  $f(a)$  is missing — **or the match**, if  $f(a)$  is there and sits at the wrong height

**infinite**

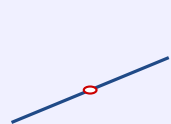
**the limit** — both sides run to the same infinity ( $\pm\infty$ ), or they disagree (DNE). Either way there is no number

**jump**

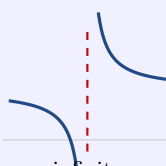
**the limit** —  $\text{LHL} \neq \text{RHL}$ , so DNE

**essential** (oscillatory)

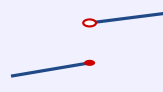
**the limit** — *neither* one-sided limit exists. Nothing to jump between and nothing to fill in



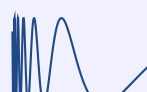
removable



infinite



jump



essential

## FAMILIAR FUNCTIONS ARE CONTINUOUS EVERYWHERE ON THEIR DOMAIN

polynomial · root · trig · rational · exponential · logarithmic

**So for any standard toolkit function, look for places outside the domain of the function — that is where continuity breaks.** You find them by *factoring*, not by taking a limit at every point.

And the list is closed: if  $f$  and  $g$  are continuous at  $a$ , so are  $f + g$   $f - g$   $cf$   $fg$   $\frac{f}{g}$  (if  $g(a) \neq 0$ )  $f \circ g$ . **Anything you build out of familiar functions with those six is continuous everywhere on its own domain too.**

## INTERVALS OF CONTINUITY

At the edge of a domain the two-sided test **cannot be run** — there is nothing on the other side. Demanding it would call every root function broken at its own edge.

$$\text{right-continuous at } a : \lim_{x \rightarrow a^+} f(x) = f(a)$$

Mirror it for a left endpoint.  $\sqrt{x-1}$  is continuous on  $[1, 2]$ .

This is also what decides a bracket: an interval closes with  $[]$  at an endpoint the curve actually *arrives at* from inside, and with  $()$  where it does not.

## INTERMEDIATE VALUE THEOREM

$$f \text{ cont. on } [a, b], N \text{ between } f(a) \text{ and } f(b) \Rightarrow \exists c \in (a, b), f(c) = N$$

“it **MUST TRACE THRU**”

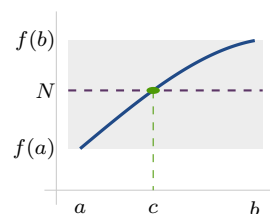
Draw a continuous function at height  $f(a)$  when  $x = a$  and  $f(b)$  when  $x = b$ . To get from one to the other it has to **sweep through every height in between**.

The move — to show two things must be equal:

$$f(x) = \text{thing1}(x) - \text{thing2}(x)$$

then show  $f$  has a **root**. A root of the difference is a point where the two agree.

It gives **existence only**. Never where. Never how many. And “between” is order-free —  $f(b)$  may sit below  $f(a)$ .



every height in the band gets hit — at least once

## WATCH OUT

✗ ~~a corner is a discontinuity~~ → continuity is not smoothness —  $|2x - 1|$  is continuous at  $\frac{1}{2}$

✗ ~~removable means  $f(a)$  is missing~~ → it can be there and simply *wrong* — the limit is the honest answer

✗ ~~three kinds is a theorem~~ → it is a list of the common ones —  $\sin(1/x)$  at 0 is the fourth

✗ ~~you can substitute anywhere~~ → only on the domain — so look outside the domain, that is where it breaks

✗ ~~the theorem hands you the  $c$~~  → existence only; never where, never how many

✗ ~~IVT needs  $f(a) \leq N \leq f(b)$~~  → *between*, either way round —  $f(b)$  may be the smaller one